



Artificial Intelligence Adoption in Green Micro, Small, and Medium Enterprises: Lessons From India for Advancing the Green Economy in Afghanistan

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Abstract: Artificial intelligence is a transformative enabler of sustainable business practices, offering substantial opportunities to enhance the environmental and operational performance of micro, small, and medium enterprises. Despite growing global interest in artificial intelligence-driven sustainability, limited attention has been directed toward its application in resource-constrained and conflict-affected economies. This study aims to examine the role of artificial intelligence in facilitating sustainable practices among green micro, small, and medium enterprises and to evaluate the applicability of successful artificial intelligence adoption experiences from India to Afghanistan. A qualitative research design based on a structured literature review was employed, and relevant secondary data were systematically collected from peer-reviewed publications indexed in Scopus, Science Direct, and Google Scholar. The reviewed evidence demonstrates that artificial intelligence can significantly improve resource efficiency, energy management, waste reduction, supply chain optimization, environmental monitoring, green marketing, and eco-innovation. The Indian experience indicates that successful artificial intelligence adoption has been facilitated by supportive policy frameworks, expanding digital infrastructure, and continuous capacity development for micro, small, and medium enterprises. In contrast, artificial intelligence adoption in Afghanistan is constrained by inadequate digital infrastructure, limited technological readiness, and insufficient financial resources. Nevertheless, the findings suggest that these barriers can be progressively mitigated through context-specific implementation strategies. It is therefore concluded that artificial intelligence can serve as a practical catalyst for accelerating sustainable micro, small, and medium enterprise transformation and promoting green economic development in Afghanistan when integrated with coordinated investments in digital infrastructure, technical capacity development, supportive policy frameworks, and financial support mechanisms. By synthesizing existing knowledge and translating successful practices from India into the Afghan context, this study contributes to the growing literature on artificial intelligence-driven sustainable entrepreneurship and provides evidence-based policy recommendations for fostering environmentally sustainable and economically resilient micro, small, and medium enterprises in developing economies.

Keywords: Artificial intelligence; Green micro, small, and medium enterprises; Sustainable development; Green innovation; Green economy

1 Introduction

With the rapid development of the digital environment, despite other changes, artificial intelligence is on the rise due to its unique capabilities. Artificial intelligence has attracted widespread attention across society and all sectors due to its efficiency and speed in various fields [1]. It is extensively recognized as a core driver of the Fourth Industrial Revolution, reflecting its profound influence on modern technological and economic systems [2]. Artificial intelligence is the capacity of machines to simulate human intelligence using technologies such as image recognition, natural language processing, data mining, deep learning, and machine learning. It enhances the effectiveness and caliber of decision-making procedures by fusing computer science with enormous datasets [3, 4]. Additionally, by combining artificial intelligence with big data, businesses can turn disparate data into actionable insights that accelerate and improve decision-making [5, 6]. According to earlier research, adopting artificial intelligence improves

corporate performance by reducing expenses, enhancing forecasting, streamlining processes, boosting productivity through automation, and encouraging innovation [7]. Moreover, artificial intelligence readiness helps micro, small, and medium enterprises make better decisions, boost operational efficiency, empower employees, and foster socially responsible and ethical business practices, ultimately improving overall business performance [8], optimizing operations, increasing productivity through automation, and fostering innovation [9]. Because of this, companies are spending more on artificial intelligence to boost their competitiveness, despite the fact that adoption still faces substantial obstacles [10].

In this regard, micro, small, and medium enterprises are a crucial area for artificial intelligence applications to achieve both environmental and economic objectives. Those enterprises in India are progressively integrating artificial intelligence to enhance supply chain efficiency, productivity, and customer engagement through the use of digital tools and supportive laws [11]. Artificial intelligence has the potential to contribute to green micro, small, and medium enterprises in reducing waste, enhancing resource utilization, managing energy use, and encouraging sustainable business practices, offering valuable perspectives into sustainable development [12, 13]. The Afghani economy is very much reliant on micro, small and medium enterprises, these enterprises contribute to the economic recovery in Afghanistan, but are severely constrained by deep economic shocks, inadequate infrastructure, and lack of access to credit and bank guarantees. The necessity for targeted and cost-effective interventions is especially important in this context, as the uptake of sustainable practices, and overall private sector growth, are further hampered by high financial hurdles and weak institutional support [14]. Therefore, large-scale adoption of artificial intelligence is not immediately possible. Instead, specific and selective applications offer a more realistic path.

It would be better for Afghanistan to have an Indian model that includes joint digital services and pilot projects in a cluster and subsidized green refurbishments. Empowering micro, small, and medium enterprises with artificial intelligence and related policies, digital infrastructure, human resources, and financial support can be critical for boosting their competitiveness and sustainable growth, as suggested by similar findings in developing economies [12, 15]. Afghanistan can also prioritize simple artificial intelligence tools that deliver immediate benefits, such as demand forecasting, maintenance planning, and energy usage monitoring, as these approaches require less infrastructure than cutting-edge artificial intelligence systems. The policy lesson is straightforward: When finance, training, and demonstration projects are combined, green transformation has the biggest impact on micro, small, and medium-sized firms [12]. However, micro, small, and medium enterprises face multiple barriers to adopting artificial intelligence-driven green practices, including high costs, weak digital infrastructure, limited technical skills, and uncertainty regarding implementation [16]. These difficulties are especially important in weak economies with low levels of technical preparedness. Because of this, even while artificial intelligence has been successful in promoting green habits in nations like India, its direct applicability in Afghanistan is still somewhat limited.

This creates a critical research gap on how to adapt artificial intelligence-based green practices to resource-constrained environments. Without low-cost, context-appropriate artificial intelligence solutions, the transition to sustainable micro, small, and medium-sized enterprises in Afghanistan is likely to face implementation challenges [16]. Accordingly, this study intends to (i) examine the role of artificial intelligence in promoting green practices among micro, small, and medium enterprises, with reference to selected initiatives in India, and (ii) analyze the applicability of artificial intelligence-driven practices from India for advancing green economy practices among micro, small, and medium enterprises in Afghanistan.

2 Literature Review

2.1 Literature Review Summary

Table 1 provides a synthesis of the existing literature on artificial intelligence applications in micro, small, and medium enterprises.

Table 1. Literature review summary

Ref.	Research Focus	Major Findings	Relevance to Afghanistan
[1]	Artificial intelligence and firms' performance	Implementing artificial intelligence enhances companies' strong, long-term performance.	Encourages Afghanistan's transition to green micro, small, and medium enterprises.
[7]	Challenges of sustainable manufacturing adoption in businesses	Micro, small, and medium enterprises encounter organizational, technical, and financial obstacles to long-term change.	Provide policy insight and useful information for low-income countries such as Afghanistan.

[8]	Artificial intelligence technology readiness for social sustainability and business ethics	Business ethics plays an important mediation role between the readiness of an organization to embrace artificial intelligence and its social sustainability.	Encourages Afghanistan's transition to green micro, small, and medium enterprises.
[9]	Artificial intelligence, business expansion, and new product development	Artificial intelligence is one of the new technologies that can support product innovation and growth.	Offering policy and useful information for Afghanistan.
[16]	Influential factors that support the sustainable growth of micro, small, and medium enterprises	Artificial intelligence promotes sustainable digital transformation, lowers waste, and boosts the efficiency of micro, small, and medium enterprises.	Highlight the needs for sustainable growth of green micro, small, and medium enterprises in Afghanistan.
[17]	Artificial intelligence for sustainability	Artificial intelligence greatly enhances the operational and financial performance of micro, small, and medium enterprises.	Encourage transition to green micro, small, and medium enterprises in Afghanistan.
[18]	Artificial intelligence for micro, small, and medium enterprises: sustainable development	Despite infrastructure and financial obstacles, artificial intelligence increases productivity and resource efficiency.	Emphasizes the significance of technological preparedness and digital infrastructure for Afghan micro, small, and medium enterprises.
[19]	Artificial intelligence use in micro, small, and medium enterprises	Suggests an artificial intelligence–Internet of Things framework for micro, small, and medium enterprises to adopt Industry 4.0.	Useful for strengthening the sustainability of micro, small, and medium enterprises in Afghanistan.
[20]	Digital marketing strategy for sustainable performance of micro, small and medium enterprises	The study finds significant gaps in the literature on the sustainability of micro, small, and medium enterprises.	Encourages Afghanistan's transition to green micro, small, and medium enterprises.
[21]	Green economy sustainability for micro, small, and medium enterprises	Green innovation and the sustainability of micro, small, and medium enterprises are strengthened by self-efficacy and work enthusiasm.	Encourages micro, small, and medium enterprises in Afghanistan to adopt sustainable and eco-friendly business practices.
[22]	Green finance and artificial intelligence in India	The policy and technological evolution and green finance are facilitating the use of artificial intelligence in India.	Encourages Afghan micro, small, and medium enterprises to adopt sustainable and eco-friendly business practices.
[23]	Potential and challenges of green micro, small, and medium enterprises through artificial intelligence	Despite research and technical hurdles, green artificial intelligence holds great promise for sustainability.	Encourages Afghanistan's transition to green micro, small, and medium enterprises.
[24]	Effects of artificial intelligence on economic, social, and business performance	Innovation and artificial intelligence technology support sustainability in the economy, society, and environment.	Supports the transition of micro, small, and medium enterprises in Afghanistan towards green and sustainable business practices.

2.2 Reviews of Previous Studies

Artificial intelligence has developed into a game-changing tool for enhancing micro, small, and medium enterprises' sustainability and performance. Studies show that artificial intelligence adoption improves resource utilization, reduces operational costs, and strengthens supply chain and inventory management systems, leading to significant gains in efficiency and productivity [17]. Additionally, micro, small, and medium enterprises may optimize logistics and lessen their environmental impact thanks to artificial intelligence-enabled big data analytics, which is essential for enhancing decision-making and promoting efficient green supply chain management [25]. Furthermore, artificial

intelligence is recognized as a main motor of the sustainability of micro, small, and medium enterprises, contributing to improved operational performance and reduced environmental footprints; however, challenges such as high initial costs, lack of infrastructure, and weak policy support continue to limit widespread adoption [18]. From a technical standpoint, sophisticated outlines that pool artificial intelligence with block chain, cloud computing, and the Internet of Things have been suggested to help micro, small, and medium enterprises make the shift to Industry 4.0. Predictive maintenance, generative design, and effective supply chain operations are made possible by these technologies, but micro, small, and medium enterprises are still susceptible to cybersecurity threats and resource constraints, necessitating government assistance and worker training for successful adoption [19]. In addition, artificial intelligence has become a powerful force in green innovation, helping to generate predictive analytics, efficient utilization of resources, and creation of eco-friendly products and processes. By leveraging artificial intelligence alongside cutting-edge digital technologies, companies can optimize their operations, boost resilience, and drive data-informed sustainable innovation at a faster pace [26].

Micro, small, and medium enterprises also contribute to social well-being, environmental preservation, and economic growth, all of which are critical to attaining sustainable development. In addition to highlighting their significance across a range of sustainability dimensions, such as environmental performance, green management techniques, and sociocultural issues, bibliometric and systematic reviews also identify gaps in theoretical development and in global comparative research [27]. Furthermore, empirical research shows that eco-entrepreneurship, collaborative networks and sustainable innovation significantly enhance the performance of micro, small, and medium enterprises in the context of the circular economy. However, the alignment of financial support mechanisms with the capabilities of micro, small, and medium enterprises determines their effectiveness, underscoring the need for tailored policy interventions [19]. Additionally, the micro, small and medium enterprises are aware of the importance of environment due to the green economic incentives which further encourages them to take part in the practices of circular economy like internal environmental management, eco-design and asset recovery by the companies to sustain the development of businesses [28]. Eco-innovation and green entrepreneurship are seen as key drivers of sustainable business practices at the firm level, with green innovation having a significant direct impact and being reinforced by regulatory incentives and training [29]. Furthermore, despite barriers related to infrastructure, costs, and digital skills, digital transformation, particularly through digital marketing, improves the sustainability of micro, small, and medium enterprises by increasing marketplace access, consumer engagement, and competitiveness [20].

Additionally, micro, small, and medium enterprises' readiness for green entrepreneurship is greatly enhanced by institutional support and environmental knowledge; sector-specific variations necessitate stakeholder collaboration and customized policy measures, green innovation and long-term sustainability commitment are also influenced by psychological aspects, including self-efficacy and work-related enthusiasm, underscoring the need for capacity-building programs [21]. Furthermore, the idea of green artificial intelligence emphasizes the necessity of lowering artificial intelligence's negative environmental effects through sustainable infrastructure, energy-efficient models, and optimized algorithms [30].

Furthermore, artificial intelligence is essential for advancing green and sustainable practices across all organizational functions. While supply chain management, despite issues of cost and integration complexity, improves efficiency and reduces environmental impacts, green marketing uses artificial intelligence to create tailored strategies that encourage ethical consumption and increase transparency [31, 32]. Furthermore, although strict digital tactics may limit their effectiveness in achieving sustainability goals, artificial intelligence-based technology systems promote economic and socio-environmental innovation [24]. As artificial intelligence increases the impact of environmental technologies and renewable energy, empirical evidence increasingly displays that green technology innovation and artificial intelligence are crucial for achieving long-term sustainable development [33]. The concept of green artificial intelligence further emphasizes two key dimensions: green by artificial intelligence, which applies artificial intelligence to resolve ecological challenges, and green in artificial intelligence, which emphasizes reducing the ecological footprint of artificial intelligence systems themselves. This dual approach is essential for aligning technological advancement through sustainability goals [34]. In emerging economies such as India, artificial intelligence has been identified as a critical driver of efficiency in renewable energy, waste management, and precision agriculture, supporting the transition toward sustainable development despite challenges related to cost and infrastructure. Additionally, studies based on the technology organization environment framework indicate that artificial intelligence readiness significantly enhances social sustainability and ethical business practices in micro, small, and medium enterprises, particularly in developing countries [8].

In the Indian framework, the adoption of artificial intelligence for sustainability is influenced by structural, technological, and human factors. The use of artificial intelligence in the context of sustainability in India is supported by supportive policy frameworks, green finance initiatives, and technological advancement. The use of artificial intelligence in the context of sustainability in India is supported by supportive policy frameworks, green finance initiatives, and technological advancement. The potential of artificial intelligence in green investments in decision making, transparency and risk management, government policies and innovative financial products

make sustainable development and resilience to climate change more attractive [22]. At the same time, various green artificial intelligence initiatives focusing on model efficiency, cloud optimization, and carbon reduction have emerged to address the environmental challenges associated with artificial intelligence technologies [23]. Despite these advances, micro, small, and medium enterprises still face relevant barriers to overcome when implementing artificial intelligence and moving towards sustainability. Limited organizational resources, financial risks, absence of technical knowledge, and uncertainty about the benefits of green practices are key barriers that require focused solutions such as financial incentives and educational initiatives [35]. Although frameworks that emphasize alignment between technology, organizational culture, workforce training, and financial investment offer pathways for effective deployment, organizational and technological constraints further hinder artificial intelligence adoption [16]. In addition, internal variables such as innovation orientation, management support, and digital competencies are essential to overcome barriers to adoption and improve the competitiveness of micro, small, and medium enterprises [36].

The literature review shows that artificial intelligence has emerged as a critical factor in the sustainable growth of micro, small, and medium enterprises, contributing to improved operational efficiency, resource utilization, green innovation, and environmental performance. A few previous studies have reliably identified key factors such as digital infrastructure, financial resources, institutional readiness, and responsible artificial intelligence practices as crucial in achieving success in implementation. Yet the current literature is predominantly focused on technologically advanced or relatively stable developing economies and primarily examines the general benefits and barriers to artificial intelligence adoption in micro, small, and medium enterprises. Little consideration has been paid to the potential adaptation of these successful artificial intelligence-powered green practices in fragile and resource-constrained economies, with varying institutional and technological settings.

While the existing literature offers significant insights into the application of artificial intelligence to the sustainability of micro, small, and medium enterprises, the focus has been limited to contexts with no resource scarcity or fragility. Past literature has offered many insights into the use of artificial intelligence for the sustainability of micro, small, and medium enterprises, but focuses little on fragile, resource-constrained contexts like Afghanistan. The results of most studies are not readily applicable to economies with structural problems that hinder technological progress and institutional capacity. The lack of robust digital infrastructure, technical skills, financial resources, and policy support, among other factors, continues to hinder the adoption of artificial intelligence-powered green practices among micro, small, and medium enterprises in Afghanistan. Moreover, there is no research exploring what is possible when the successful implementation of artificial intelligence-based green measures is translated into Afghanistan's socio-economic and institutional landscape. In this context, this structured literature review aims to fill this gap by summarizing the current body of evidence and drawing practical lessons from India's experience to inform context-specific ways to support artificial intelligence-powered green micro, small, and medium enterprises and the green economy in Afghanistan.

3 Research Methodology

3.1 Literature Search Strategy

This study employs a qualitative research methodology, grounded in a structured literature review, to examine the role of artificial intelligence in promoting green practices among micro, small, and medium enterprises and to explore how India's experience can inform green economy practices in Afghanistan. The literature contains a number of structured literature review techniques that are made up of numerous rigorous processes since they serve as a means of both organizing existing publications and advancing knowledge in a variety of study domains [37]. Relevant literature published between 2020 and 2025 was systematically collected using Scopus, Web of Science, Google Scholar, and Springer Link. In addition, relevant publications from reputable publishers, including journals published by the Multidisciplinary Digital Publishing Institute (MDPI). The data retrieval process employed Boolean operators (AND/OR) and followed advanced search queries using specific and relevant keywords to ensure the inclusion of studies aligned with the research focus ("artificial intelligence" OR "green micro, small, and medium enterprises" OR "green economy" OR "small and medium enterprises" OR "sustainable business" OR "artificial intelligence adoption" OR "green practices" OR "green technology" OR "green innovation" OR "India" AND "Afghanistan").

3.2 Study Selection Criteria

After searching the literature, the selected publications were screened using the set inclusion and exclusion criteria to include only studies directly relevant to the objectives of this review (Table 2).

3.3 Research Design

The study's descriptive and exploratory design aims to analyze current knowledge and investigate how artificial intelligence-driven green practices might be applied in various economic scenarios. The study is organized around two main pillars: (i) analyzing how artificial intelligence may improve micro, small, and medium enterprises sustainability, and (ii) assessing how Afghanistan can learn from India.

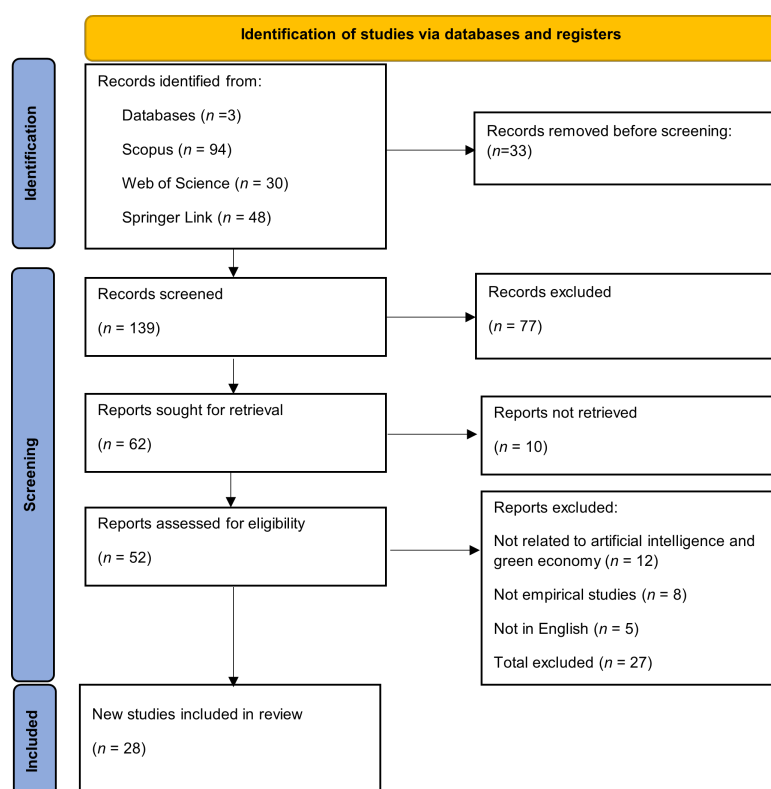
Table 2. Inclusion criteria and exclusion criteria

Inclusion Criteria	Exclusion Criteria
Articles that specifically discuss micro, small, and medium enterprises operating in developing economies' adoption and application of artificial intelligence (with an emphasis on Afghanistan and India).	Duplicate publications identified across different databases.
Focused on publications published between 2020 and 2025, reflecting the rapid development of artificial intelligence technologies and the growing scholarly interest in green micro, small, and medium enterprises.	Studies focusing exclusively on large corporations or multinational enterprises rather than micro, small, and medium enterprises.
Studies focusing on artificial intelligence in micro, small, and medium enterprises, green micro, small, and medium enterprises, artificial intelligence for green practices, India's experience, challenges, and barriers.	Publications unrelated to artificial intelligence, micro, small, and medium enterprises, or sustainable development.
Peer-reviewed journal articles, selected conference papers, and international reports relevant to the research objectives, and publications written in English.	Publications not available in English.

The study employs a contextual comparative approach with India as a reference country to discover transferable green practices in artificial intelligence in Afghanistan. This analysis involves a thematic categorization of the literature along key areas, including

- Artificial intelligence in micro, small, and medium enterprises
- Green micro, small, and medium enterprises and sustainability
- Artificial intelligence for green practices
- India's experience
- Challenges and barriers

Figure 1, which gives a clear summary of the review approach used in this study, depicts the entire literature identification, screening, eligibility evaluation, and final study selection process.

**Figure 1.** Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of study selection

4 Results, Data Analysis and Interpretation

The literature review reveals that artificial intelligence has emerged as a key enabler of green transformation in micro, small, and medium enterprises. In short, the examined studies show that the use of artificial intelligence generally improves operational efficiency, resource utilization, decision-making, and environmental effects due to machine learning, big data, predictive maintenance, and smart supply chain management and process automation. The thematic analysis also shows that artificial intelligence achieves sustainability by reducing waste, improving energy efficiency, accelerating eco-innovation, and promoting environmentally friendly business practices. Findings show that artificial intelligence does not merely represent a technology but a strategy that can support the long-term transition to a green economy. The evidence also shows that successful adoption of artificial intelligence relies on more than just available technology. According to the studies considered, organizational readiness, digital skills, financial investment, pro-innovation government policy, and institutional capacity determine if artificial intelligence can produce sustainable outcomes. This is important for developing economies because technological adoption occurs in an ecosystem of capabilities rather than technology per se. Artificial intelligence ought to be seen as a component of enhanced digital transformation that involves technology, people, governance and sustainable business practices.

India is identified as an important case study in understanding how artificial intelligence can help develop green micro, small, and medium enterprises. India's experience indicates that coordinated government initiatives, expansion of digital infrastructure, increasing technology awareness, financial support mechanisms, and continuing investment for innovation build favorable conditions for the adoption of artificial intelligence. The micro, small, and medium enterprises in India have used artificial intelligence in renewable energy management, precision agriculture, waste management, digital marketing, intelligent supply chain systems, etc. This development has delivered improved productivity, resource efficiency, and environmental performance. These initiatives are not just isolated technological successes. They can illustrate how supportive institutions and coordinated policies can help achieve sustainable digital transformation. Nevertheless, the results also reveal that India's experience should not be seen as a model that can be transferred to Afghanistan. Significant differences exist between the two countries in terms of digital infrastructure, institutional capacity, financial resources, technical expertise, and policy implementation. As such, the key takeaway from India is not to chase advanced artificial intelligence but rather the need to establish the institutional and technological foundations to facilitate a gradual adoption of artificial intelligence.

This thematic analysis also highlights key persistent barriers to artificial intelligence adoption among micro, small, and medium enterprises. The major barriers to sustainable artificial intelligence implementation consistently reported across the reviewed studies are high upfront costs, poor digital infrastructure, limited technical skills and expertise, financial resource limitations, organizational resistance to change, and ineffective policy support. These barriers not only hold true in many developing countries for micro, small, and medium enterprises, but the challenge becomes way bigger in fragile economies, as their institutional capacity and technological readiness are still quite low. Hence, the results imply that sustainable transition is not possible with technological innovation alone, but requires simultaneous investments in infrastructure, skills and supportive governance. The results of the study are important for Afghanistan. According to the analyzed data, Afghanistan lacks many of the elements that have made it easier for India to employ artificial intelligence, such as proper digital infrastructure, skilled human resources, financial resources, and encouraging regulatory policies. Therefore, the idea of fast and broad adoption of artificial intelligence is unfeasible. Rather, Afghanistan's strategy should be a step-by-step one that builds the capacity of institutions and gradually improves its technological capacity, depending on the context.

In terms of implementation, the literature review shows several potential short-term applications of artificial intelligence that can be used in Afghanistan. As an alternative to investing in high-end artificial intelligence technologies, the policymakers and micro, small, and medium enterprises can focus on low-cost digital solutions like artificial intelligence-powered demand forecasting, inventory management, energy monitoring, predictive maintenance, digital marketing, and simple supply chain optimization. The technologies involve relatively small investments in funds, but offer measurable gains in operational efficiency and resource utilization. This incremental approach would enable Afghan micro, small, and medium enterprises to familiarize themselves with artificial intelligence in a tangible way, minimize implementation risks, and foster their confidence in expanding technology.

The results also show that policy coordination is more vital than technological investment, if the implementation is to be successful. The focus should thus be on improving digital infrastructure, increasing internet connectivity, building digital literacy and technical training, providing access to affordable finance options for micro, small, and medium enterprises, and establishing policies that promote responsible artificial intelligence use and green innovation. Furthermore, the cooperation between government institutions, universities, private companies, and international development organizations can help to speed up the transfer of knowledge, development of technical capability, and investment in sustainable technology. Such policy measures are allied with the evidence that has been consolidated from the reviewed studies and feasible pathways for facilitating Afghanistan's gradual transition towards artificial intelligence-enabled green micro, small, and medium enterprises.

Overall, the structured literature review shows that artificial intelligence has significant potential for the sustainable

development of micro, small, and medium enterprises, and its impact depends on the interaction among technological capability, institutional readiness, organizational capacity, and supportive public policy. The experience gained in India is a lesson rather than a blueprint to be replicated. A slow, adaptive, and resource-sensitive path that emphasizes affordable technologies, institutional development, and human capacity building is likely to yield more sustainable results in Afghanistan. Based on the evidence reviewed, the conclusion is that artificial intelligence can be an important catalyst for the transition to a green economy in Afghanistan when applied through realistic, context-specific policies that reflect current developmental conditions. Figure 2 shows the conceptual framework for artificial intelligence-driven green micro, small, and medium enterprises in Afghanistan.

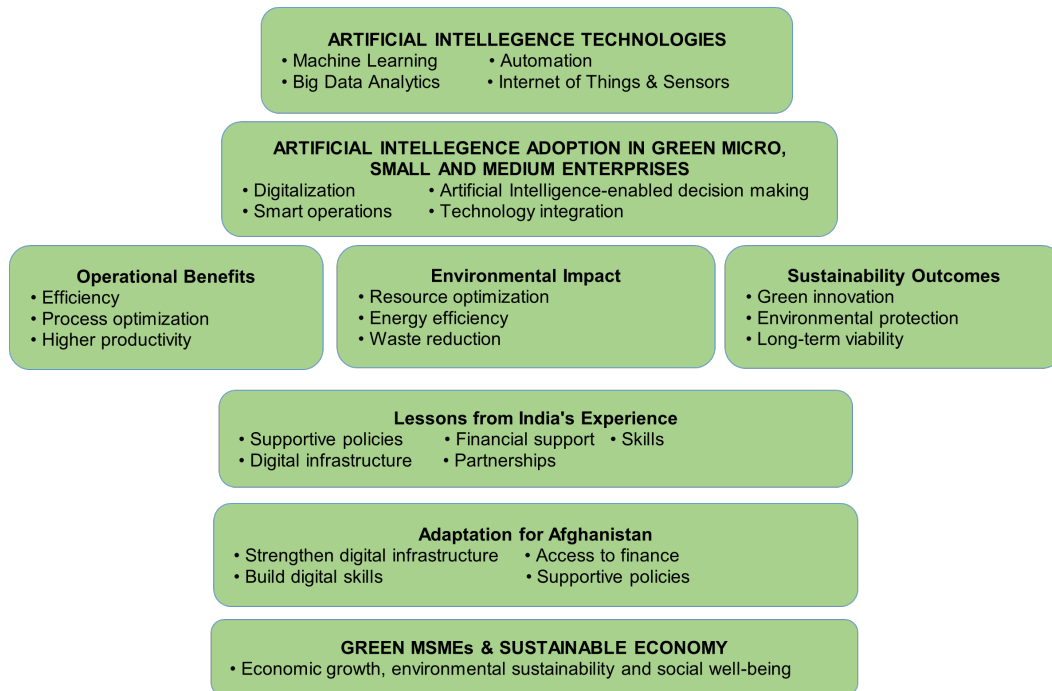


Figure 2. Conceptual framework for artificial intelligence-driven green micro, small, and medium enterprises in Afghanistan

5 Conclusion and Policy Recommendations

This study examined how artificial intelligence can support micro, small, and medium enterprises in implementing sustainable and eco-friendly practices. The study focused on applying lessons learned from India to Afghanistan. The results, founded on a systematic review of the literature, suggest that artificial intelligence has significant potential to improve operational effectiveness, maximize resource utilization, and reduce environmental impacts in the operations of micro, small, and medium enterprises. Businesses can adopt more sustainable production and management practices, which will help achieve the larger goals of a green economy, due to artificial intelligence-based solutions such as data analytics, automation, and smart supply chain systems. The analysis also shows that India has made great strides in integrating artificial intelligence into the development of micro, small, and medium enterprises, due to supportive regulations, digital infrastructure, and growing adoption of the technology. Indian micro, small, and medium enterprises have effectively used artificial intelligence in areas such as digital marketing, waste management, and renewable energy, proving that a coordinated strategy that combines technology, institutional support, and human talent can drive long-term change. On the other hand, Afghanistan faces significant institutional and structural barriers, such as insufficient digital infrastructure, low levels of technical abilities, financial constraints, and fragile policy frameworks, which hinder the adoption of green artificial intelligence-based practices.

Despite these drawbacks, this analysis suggests that Afghanistan can benefit greatly from customizing the Indian experience. Afghanistan's shift towards green artificial intelligence-based micro, small, and medium enterprises should be seen as a progressive and adaptive process that aligns with local institutional capacity and economic realities, rather than as a direct copy of foreign models. To achieve sustainable development in the micro, small, and medium enterprise sector, the gap between the potential of the technology and actual application still needs to be bridged. From a policy perspective, this research yields several strategic recommendations. First, to enable the adoption of artificial intelligence, it is essential that digital infrastructure, including internet connectivity and access to affordable technological tools, be developed and strengthened. Second, to reduce the financial burden of artificial

intelligence implementation for micro, small, and medium enterprises, governments and policymakers should put in place certain financial incentives, such as subsidies, grants, and low-interest loans. Third, capacity-building initiatives such as technical training programs and digital literacy campaigns are crucial to equip workers and businesses with the skills needed to effectively use artificial intelligence technology.

A supportive regulatory framework that promotes innovation while ensuring the ethical and long-term application of artificial intelligence is also essential. In order to increase knowledge transfer, technology diffusion, and investment in green initiatives, policymakers should also encourage collaboration between government agencies, businesses, and international organizations. Furthermore, by fostering an innovation-driven culture, reinforced by developing leadership and entrepreneurial drive, the adoption of sustainable practices in micro, small, and medium enterprises can be greatly enhanced. In conclusion, artificial intelligence presents a powerful opportunity to transform micro, small, and medium enterprises into key drivers of the green economy. While India provides valuable lessons in leveraging artificial intelligence for sustainable development, Afghanistan must adopt a tailored and phased approach to harness these opportunities effectively. By aligning technological advancement with policy support, institutional capacity, and local needs, Afghanistan can gradually build a resilient and sustainable sector of micro, small, and medium enterprises that contributes to long-term economic growth and environmental sustainability.

6 Limitations of the Study and Future Research

This study has its limitations as it only used secondary data and a structured literature review instead of collecting primary data from the micro, small, and medium enterprises or policymakers. Furthermore, the scarcity of research specific to Afghanistan necessitated increased research on Afghanistan's counterpart, India, and other developing economies, thereby limiting the applicability of some results. Empirical/survey research, case studies, and comparative studies of artificial intelligence-driven green micro, small, and medium enterprises in other developing nations should confirm these findings and offer more localized evidence for the effect of artificial intelligence on green micro, small, and medium enterprises in Afghanistan.

Author Contributions

Conceptualization, A.W.G. and K.S.; methodology, A.W.G.; software, A.W.G.; validation, A.W.G. and K.S.; formal analysis, A.W.G.; investigation, A.W.G.; resources, A.W.G.; data curation, A.W.G.; writing—original draft preparation, A.W.G.; writing—review and editing, A.W.G. and K.S.; visualization, A.W.G.; supervision, K.S.; project administration, A.W.G. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data supporting our research results were obtained from published literature, which is freely available and can be found within the article via the cited references. No primary data have been collected.

Conflicts of Interest

The authors declare no conflicts of interest.

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